Lab 7: Fuzzing for Input Validation Bugs

Name: E Bhuvan
SRN: PES1UG24CS568
Class: 5J

Step 2: Buggy Application

```
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$ python parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 09:30-11:45
Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): car
Hourly Rate: 30

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): ka01ab1234
Formatted Plate: KA 01 AB 1234
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$
```

Valid input

```
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$ python parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 09:30-11:45
Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): rocket
Traceback (most recent call last):
  File "/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py", line 33, in <module>
    main()
  File "/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py", line 24, in main
    result = hourly_rate(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py", line 10, in hourly_rate
    return RATES[vehicle.strip().lower()]
           ~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^
KeyError: 'rocket'
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$
```

Invalid vehicle type

Step 4: Failing Tests

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- /home/bhu1/sem5/se/L6_L7_19_parking_fee/.venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7
plugins: hypothesis-6.170.15
collected 9 items

test_parking_fee.py::test_minutes_parked_valid_range PASSED                [ 11%]
test_parking_fee.py::test_minutes_parked_handles_arbitrary_input FAILED    [ 22%]
test_parking_fee.py::test_minutes_parked_rejects_invalid_input FAILED      [ 33%]
test_parking_fee.py::test_hourly_rate_known_vehicles PASSED                [ 44%]
test_parking_fee.py::test_hourly_rate_handles_arbitrary_input FAILED       [ 55%]
test_parking_fee.py::test_hourly_rate_rejects_unknown_vehicle FAILED       [ 66%]
test_parking_fee.py::test_format_plate_valid_plates PASSED                 [ 77%]
test_parking_fee.py::test_format_plate_handles_arbitrary_input FAILED      [ 88%]
test_parking_fee.py::test_format_plate_rejects_invalid_plate FAILED        [100%]

================================== FAILURES ===================================
E   AttributeError: 'NoneType' object has no attribute 'split'
    Failing test case: test_minutes_parked_handles_arbitrary_input(
        raw=None,
    )
/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py:4: AttributeError: 'NoneType' object has no attribute 'split'
E   Failed: DID NOT RAISE ValueError
    Failing test case: test_minutes_parked_rejects_invalid_input(
        raw='24:00-25:00',
    )
    Explanation:
        These lines were always and only run by failing test cases:
            /home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py:5
/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/test_parking_fee.py:81: Failed: DID NOT RAISE ValueError
E   AttributeError: 'NoneType' object has no attribute 'strip'
    Failing test case: test_hourly_rate_handles_arbitrary_input(
        raw=None,
    )
/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py:10: AttributeError: 'NoneType' object has no attribute 'strip'
E   KeyError: 'bus'
    Failing test case: test_hourly_rate_rejects_unknown_vehicle(
        raw='bus',  # or any other generated value
    )
/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py:10: KeyError: 'bus'
E   AttributeError: 'NoneType' object has no attribute 'upper'
    Failing test case: test_format_plate_handles_arbitrary_input(
        raw=None,
    )
/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/parking_fee.py:13: AttributeError: 'NoneType' object has no attribute 'upper'
E   Failed: DID NOT RAISE ValueError
    Failing test case: test_format_plate_rejects_invalid_plate(
        raw='KA01AB123',  # or any other generated value
    )
/home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7/test_parking_fee.py:150: Failed: DID NOT RAISE ValueError
=========================== short test summary info ============================
FAILED test_parking_fee.py::test_minutes_parked_handles_arbitrary_input - AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_parking_fee.py::test_minutes_parked_rejects_invalid_input - Failed: DID NOT RAISE ValueError
FAILED test_parking_fee.py::test_hourly_rate_handles_arbitrary_input - AttributeError: 'NoneType' object has no attribute 'strip'
FAILED test_parking_fee.py::test_hourly_rate_rejects_unknown_vehicle - KeyError: 'bus'
FAILED test_parking_fee.py::test_format_plate_handles_arbitrary_input - AttributeError: 'NoneType' object has no attribute 'upper'
FAILED test_parking_fee.py::test_format_plate_rejects_invalid_plate - Failed: DID NOT RAISE ValueError
========================= 6 failed, 3 passed in 1.54s ==========================
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$
```

pytest: 6 failed, 3 passed

Step 5: Fixed Application

```
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$ python parking_fee.py
==== Running minutes_parked ====
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Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): car
Hourly Rate: 30

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): ka01ab1234
Formatted Plate: KA 01 AB 1234
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$
```

Valid input

```
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$ python parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 25:99-10:00
Invalid input: hours must be between 00 and 23, got 25

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): rocket
Invalid input: unsupported vehicle type 'rocket'; expected one of: bike, car, truck

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): KA-1
Invalid input: number plate must be 2 letters, 2 digits, 2 letters and 4 digits (for example KA01AB1
234), got 'KA-1'
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$
```

Invalid input

Step 6: Passing Tests

```
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$ python -m pytest test_parking_fee
.py -v
================================= test session starts ==================================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- /home/bhu1/sem5/se/L6_L
7_19_parking_fee/.venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /home/bhu1/sem5/se/L6_L7_19_parking_fee/Lab 7
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test_parking_fee.py::test_format_plate_handles_arbitrary_input PASSED          [ 88%]
test_parking_fee.py::test_format_plate_rejects_invalid_plate PASSED            [100%]

================================== 9 passed in 1.63s ===================================
bhu1@bhu1-82xe:~/sem5/se/L6_L7_19_parking_fee/Lab 7$
```

pytest: 9 passed